Introduction To DevOps Assignment

DevOps Project Summary

ACEest Fitness & Gym – End-to-End CI/CD Deployment

-By

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2024tm93519

M.Tech(Software Engineering)

1. Project Overview

Developed and deployed a **Flask-based fitness web application** that includes:

- Workout tracking
- Member management
- Calorie calculator
- Gym analytics dashboard
- REST APIs + UI

This app evolved from multiple desktop versions into a **modern web application with DevOps practices**

2. Infrastructure Setup (AWS)

- Created an **EC2 instance (Linux)**
- Connected via SSH
- Installed required tools:
  - Docker
  - Jenkins
  - k3s (lightweight Kubernetes)

Configured **Security Groups**:

- Opened ports:
  - `22` → SSH
  - `8080` → Jenkins
  - `30007` → Application (NodePort)

3. Version Control (GitHub)

- Created a GitHub repository:

`introduction-to-devops-assignment-1-march2026`

- Pushed:

- Flask app (`app.py`)
- Dockerfile
- Kubernetes manifests
- Jenkinsfile

This became the **single source of truth**

4. Containerization (Docker)

- Created a **Dockerfile** for the Flask app
- Built image:

```
docker build -t aceest-app .
```

Result:

- Image stored locally in EC2
- Used by Kubernetes for deployment

5. CI/CD Pipeline (Jenkins)

- Installed Jenkins on EC2
- Created a **pipeline job**
- Connected to GitHub repo

Pipeline stages:

```
1. Checkout → Pull code from GitHub
2. Build Docker → Create Docker image
3. Deploy K8s → Apply Kubernetes manifests
```

Fully automated flow:

Code change → Jenkins build → Deployment

6. Kubernetes Deployment (k3s)

Created:

Deployment

- Runs Flask app container
- Ensures app is always running

Service (NodePort)

- Exposes app externally

```
NodePort: 30007
```

Applied using:

```
k3s kubectl apply -f deployment.yaml
k3s kubectl apply -f service.yaml
```

7. Debugging & Fixes (Key Learning)

Solved multiple real-world DevOps issues:

Issue 1: Jenkins Node Offline

Fixed by:

- Adding swap memory
- Cleaning disk space

Issue 2: Git Branch Error

```
Couldn't find revision to build
```

Fixed by:

- Using correct branch (`main` instead of `master`)

Issue 3: Docker Permission Error

```
permission denied docker.sock
```

Fixed by:

- Adding Jenkins user to Docker group

Issue 4: sudo password in Jenkins

Fixed by:

- Editing `/etc/sudoers`

```
jenkins ALL=(ALL) NOPASSWD: ALL
```

Issue 5: Kubernetes YAML path issue

Fixed by:

- Correcting file paths:

```
k3s/manifests/deployment.yaml
```

Issue 6: App not accessible

Root causes fixed:

- Security group port open
- Correct NodePort
- Correct public IP

8. Final Deployment (SUCCESS)

Application running:

- Pod: Running

- Service: NodePort exposed
- Jenkins: Pipeline successful

Access URL:

http://18.60.107.158:30007/

Successfully accessed the app in browser

9. Architecture Flow

```
GitHub → Jenkins → Docker → Kubernetes → Browser
```

1. Code pushed to GitHub
2. Jenkins pulls code
3. Docker builds image
4. Kubernetes deploys container
5. App exposed via NodePort
6. User accesses via browser

10. Key DevOps Concepts Demonstrated

- CI/CD pipeline
- Containerization (Docker)
- Orchestration (Kubernetes)
- Infrastructure setup (AWS EC2)
- Automation (Jenkins pipeline)
- Debugging real-world issues

Final Conclusion

Successfully built a **complete DevOps pipeline from scratch**:

- Developed application
- Containerized with Docker

- Automated build using Jenkins
- Deployed using Kubernetes
- Exposed to internet via AWS

Important Links Used in The Project

1. GitHub Repository (Source Code)

https://github.com/KollaNagaManasa/introduction-to-devops-assignment-1-march2026

Contains:

- Flask app
- Dockerfile
- Kubernetes manifests
- Jenkinsfile
- GitHub Actions workflow

2. GitHub Actions (CI Pipeline)

https://github.com/KollaNagaManasa/introduction-to-devops-assignment-1-march2026/actions

Shows:

- Automated test execution (pytest)
- Build status (SUCCESS)

3. Jenkins (CD Pipeline)

http://18.60.107.158:8080

Used for:

- Pipeline automation
- Docker build
- Kubernetes deployment

4. Kubernetes Deployment (CLI access)

(No direct UI link — accessed via terminal)

Commands used:

```
k3s kubectl get pods
k3s kubectl get svc
```

5. Deployed Application (Final Output)

http://18.60.146.121:30007

This is the **live application**

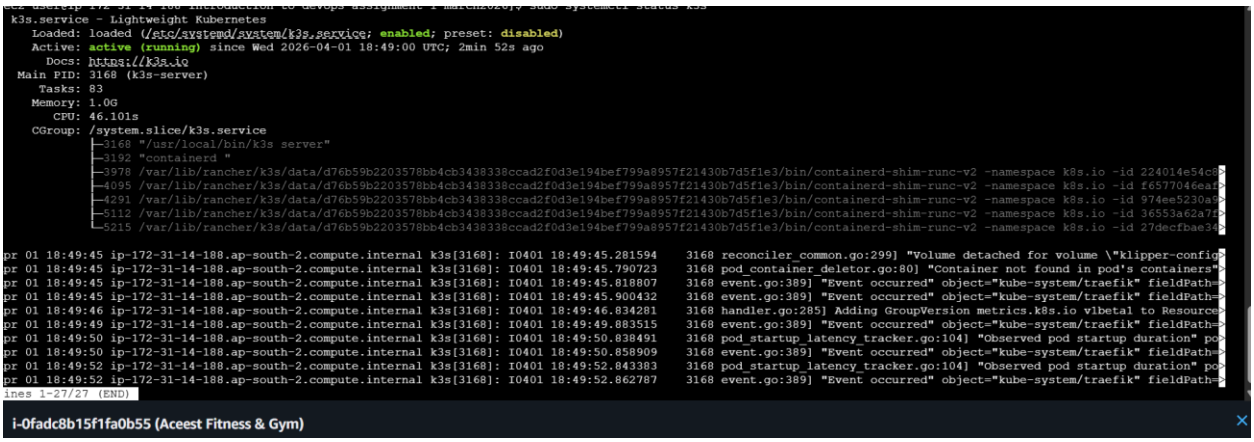
6. AWS Console (Infrastructure)

https://ap-south-2.console.aws.amazon.com/ec2

Used for:

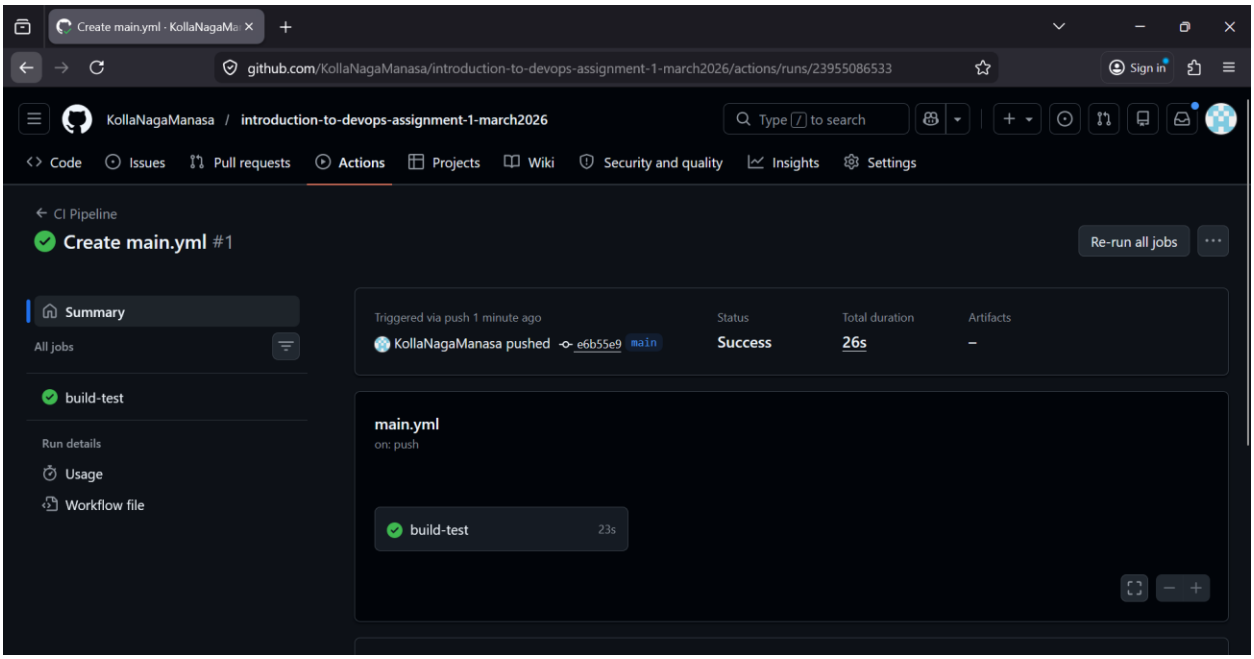
- EC2 instance
- Security groups
- Networking

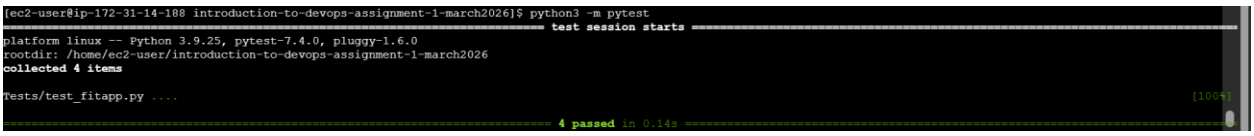
Images:

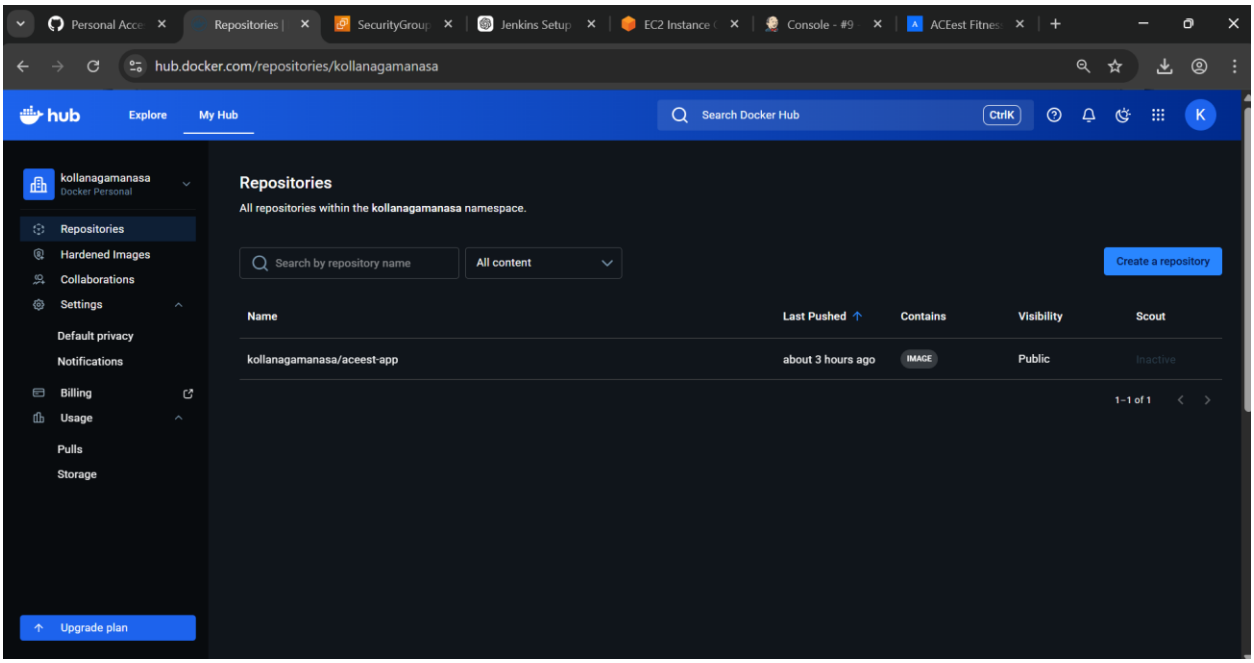


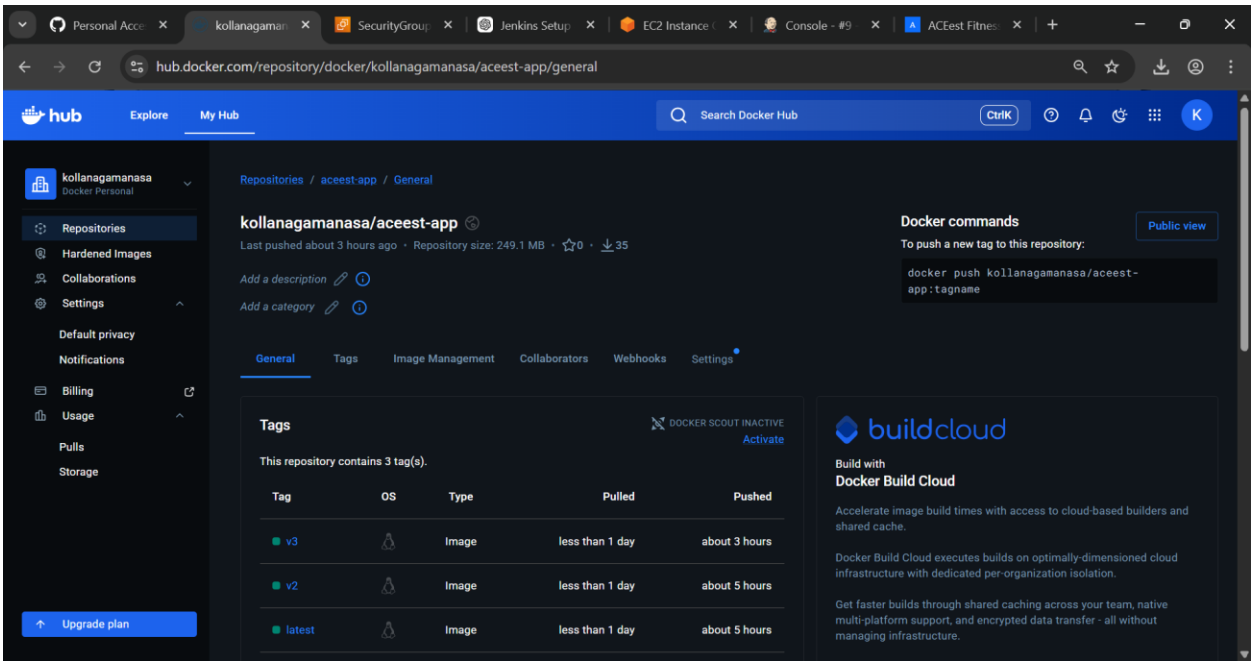
```
k3s.service - Lightweight Kubernetes
   Loaded: loaded (/etc/systemd/system/k3s.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-04-01 18:49:00 UTC; 2min 52s ago
     Docs: https://k3s.io
 Main PID: 3168 (k3s-server)
    Tasks: 83
   Memory: 1.0G
      CPU: 46.101s
   CGroup: /system.slice/k3s.service
           ├─3168 "/usr/local/bin/k3s server"
           ├─3192 "containerd "
           ├─3978 /var/lib/rancher/k3s/data/d76b59b2203578bb4cb3438338ccad2f0d3e194bef799a8957f21430b7d5f1e3/bin/containerd-shim-runc-v2 -namespace k8s.io -id 224014e54c8
           ├─4095 /var/lib/rancher/k3s/data/d76b59b2203578bb4cb3438338ccad2f0d3e194bef799a8957f21430b7d5f1e3/bin/containerd-shim-runc-v2 -namespace k8s.io -id f6577046ea
           ├─4291 /var/lib/rancher/k3s/data/d76b59b2203578bb4cb3438338ccad2f0d3e194bef799a8957f21430b7d5f1e3/bin/containerd-shim-runc-v2 -namespace k8s.io -id 974ee5230a
           ├─5112 /var/lib/rancher/k3s/data/d76b59b2203578bb4cb3438338ccad2f0d3e194bef799a8957f21430b7d5f1e3/bin/containerd-shim-runc-v2 -namespace k8s.io -id 36553a62a7
           └─5215 /var/lib/rancher/k3s/data/d76b59b2203578bb4cb3438338ccad2f0d3e194bef799a8957f21430b7d5f1e3/bin/containerd-shim-runc-v2 -namespace k8s.io -id 27decfbae3

pr 01 18:49:45 ip-172-31-14-188.ap-south-2.compute.internal k3s[3168]: I0401 18:49:45.281594    3168 reconciler_common.go:299] "Volume detached for volume \"klipper-config
pr 01 18:49:45 ip-172-31-14-188.ap-south-2.compute.internal k3s[3168]: I0401 18:49:45.790723    3168 pod_container_deletor.go:80] "Container not found in pod's containers"
pr 01 18:49:45 ip-172-31-14-188.ap-south-2.compute.internal k3s[3168]: I0401 18:49:45.818807    3168 event.go:389] "Event occurred" object="kube-system/traefik" fieldPath=
pr 01 18:49:45 ip-172-31-14-188.ap-south-2.compute.internal k3s[3168]: I0401 18:49:45.900432    3168 event.go:389] "Event occurred" object="kube-system/traefik" fieldPath=
pr 01 18:49:46 ip-172-31-14-188.ap-south-2.compute.internal k3s[3168]: I0401 18:49:46.834281    3168 handler.go:285] Adding GroupVersion metrics.k8s.io v1beta1 to Resource
pr 01 18:49:49 ip-172-31-14-188.ap-south-2.compute.internal k3s[3168]: I0401 18:49:49.883515    3168 event.go:389] "Event occurred" object="kube-system/traefik" fieldPath=
pr 01 18:49:50 ip-172-31-14-188.ap-south-2.compute.internal k3s[3168]: I0401 18:49:50.838491    3168 pod_startup_latency_tracker.go:104] "Observed pod startup duration" po
pr 01 18:49:50 ip-172-31-14-188.ap-south-2.compute.internal k3s[3168]: I0401 18:49:50.858909    3168 event.go:389] "Event occurred" object="kube-system/traefik" fieldPath=
pr 01 18:49:52 ip-172-31-14-188.ap-south-2.compute.internal k3s[3168]: I0401 18:49:52.843383    3168 pod_startup_latency_tracker.go:104] "Observed pod startup duration" po
pr 01 18:49:52 ip-172-31-14-188.ap-south-2.compute.internal k3s[3168]: I0401 18:49:52.862787    3168 event.go:389] "Event occurred" object="kube-system/traefik" fieldPath=
ines 1-27/27 (END)
```

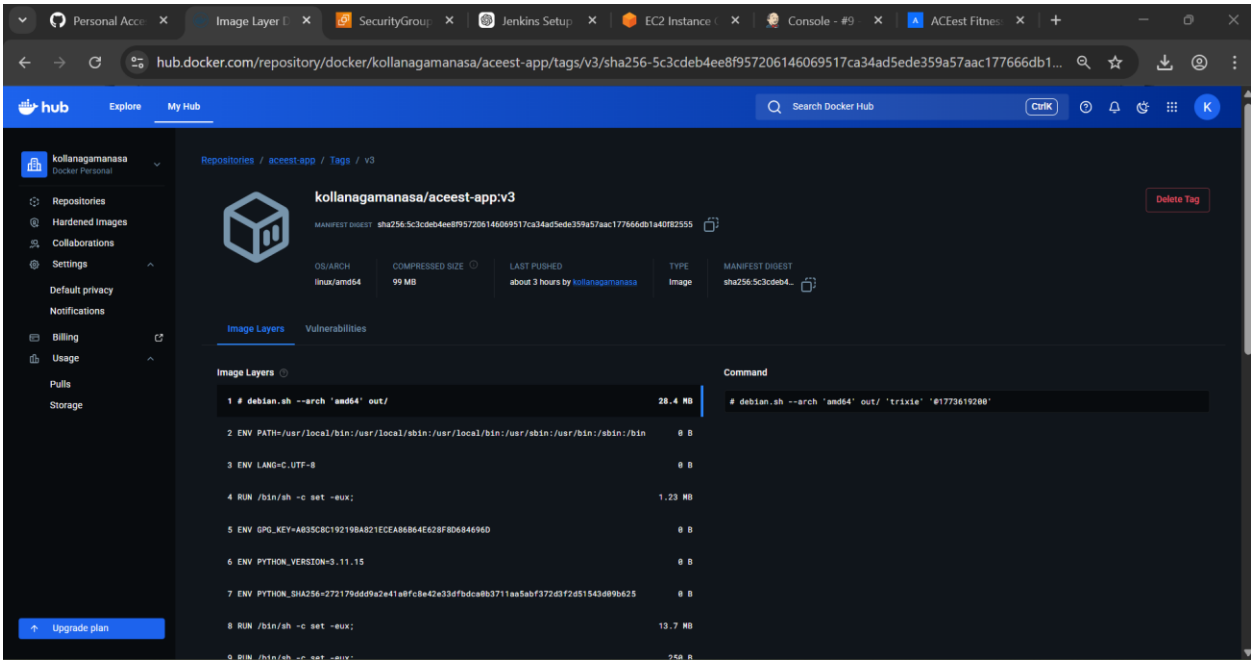
i-0fadc8b15f1fa0b55 (Aceest Fitness & Gym)


Create main.yml · KollaNagaMa
github.com/KollaNagaManasa/introduction-to-devops-assignment-1-march2026/actions/runs/23955086533
Sign in
KollaNagaManasa / introduction-to-devops-assignment-1-march2026
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
CI Pipeline
Create main.yml #1
Re-run all jobs
Summary
All jobs
build-test
Run details
Usage
Workflow file
Triggered via push 1 minute ago
KollaNagaManasa pushed e6b55e9 main
Status
Success
Total duration
26s
Artifacts
–
main.yml
on: push
build-test
23s


[ec2-user@ip-172-31-14-188 introduction-to-devops-assignment-1-march2026]$ python3 -m pytest
test session starts
platform linux -- Python 3.9.25, pytest-7.4.0, pluggy-1.6.0
rootdir: /home/ec2-user/introduction-to-devops-assignment-1-march2026
collected 4 items
Tests/test_fitapp.py ....
[100%]
4 passed in 0.14s


Personal Acce
Repositories |
SecurityGroup
Jenkins Setup
EC2 Instance
Console - #9
ACEest Fitnes
hub.docker.com/repositories/kollanagamanasa
hub
Explore
My Hub
Search Docker Hub
CtrlK
kollanagamanasa
Docker Personal
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Upgrade plan
Repositories
All repositories within the kollanagamanasa namespace.
Search by repository name
All content
Create a repository
Name
Last Pushed
Contains
Visibility
Scout
kollanagamanasa/aceest-app
about 3 hours ago
IMAGE
Public
Inactive
1-1 of 1


hub.docker.com/repository/docker/kollanagamanasa/aceest-app/general
hub
Explore
My Hub
Search Docker Hub
Ctrl K
kollanagamanasa
Docker Personal
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Upgrade plan
Repositories / aceest-app / General
kollanagamanasa/aceest-app
Last pushed about 3 hours ago • Repository size: 249.1 MB • 0 • 35
Add a description
Add a category
Docker commands
Public view
To push a new tag to this repository:
docker push kollanagamanasa/aceest-app:tagname
General
Tags
Image Management
Collaborators
Webhooks
Settings
Tags
DOCKER SCOUT INACTIVE
Activate
This repository contains 3 tag(s).
Tag
OS
Type
Pulled
Pushed
v3
Image
less than 1 day
about 3 hours
v2
Image
less than 1 day
about 5 hours
latest
Image
less than 1 day
about 5 hours
buildcloud
Build with
Docker Build Cloud
Accelerate image build times with access to cloud-based builders and shared cache.
Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.
Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.


hub.docker.com/repository/docker/kollanagamanasa/aceest-app/tags/v3/sha256-5c3cdeb4ee8f957206146069517ca34ad5ede359a57aac177666db1...
hub
Explore
My Hub
Search Docker Hub
Ctrl K
kollanagamanasa
Docker Personal
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Upgrade plan
Repositories / aceest-app / Tags / v3
kollanagamanasa/aceest-app:v3
Delete Tag
MANIFEST DIGEST sha256:5c3cdeb4ee8f957206146069517ca34ad5ede359a57aac177666db1a40f82555
OS/ARCH
linux/amd64
COMPRESSED SIZE
99 MB
LAST PUSHED
about 3 hours by kollanagamanasa
TYPE
Image
MANIFEST DIGEST
sha256:5c3cdeb4...
Image Layers
Vulnerabilities
Image Layers
1 # debian.sh --arch 'amd64' out/ 28.4 MB
2 ENV PATH=/usr/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin 0 B
3 ENV LANG=C.UTF-8 0 B
4 RUN /bin/sh -c set -eux; 1.23 MB
5 ENV GPG_KEY=A035C8C19219BA821ECEA86864E628F8D684696D 0 B
6 ENV PYTHON_VERSION=3.11.15 0 B
7 ENV PYTHON_SHA256=272179ddd9a2e41a0fc8e42e33dfbdce8b3711aa5abf372d3f2d51543d09b625 0 B
8 RUN /bin/sh -c set -eux; 13.7 MB
Command
debian.sh --arch 'amd64' out/ 'trixie' '@1773619200'

Personal Acce | Image Layer D | SecurityGroup | Jenkins Setup | EC2 Instance | Console - #9 | ACEest Fitnes

Not secure 18.60.107.158:8080/job/devops-project/9/console

Jenkins / devops-project / #9

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#9'
- Timings
- Git Build Data
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces
- Previous Build
- Next Build

Console

Download | Copy | View as plain text

```
Started by user KollaNagaManasa
Obtained Jenkinsfile from git https://github.com/KollaNagaManasa/introduction-to-devops-assignment-1-march2026.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/devops-project
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
 > git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/devops-project/.git # timeout=10
Fetching changes from the remote Git repository
 > git config remote.origin.url https://github.com/KollaNagaManasa/introduction-to-devops-assignment-1-march2026.git # timeout=10
Fetching upstream changes from https://github.com/KollaNagaManasa/introduction-to-devops-assignment-1-march2026.git
 > git --version # timeout=10
 > git --version # 'git version 2.50.1'
 > git fetch --tags --force --progress -- https://github.com/KollaNagaManasa/introduction-to-devops-assignment-1-march2026.git +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision d1ddcd2a6be6bef8e93478b40c7c22a272776142 (refs/remotes/origin/main)
 > git config core.sparsecheckout # timeout=10
 > git checkout -f d1ddcd2a6be6bef8e93478b40c7c22a272776142 # timeout=10
Commit message: "Update Jenkinsfile"
 > git rev-list --no-walk 266465b66cf423d2c15d837208cee608904cfbe5 # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
```

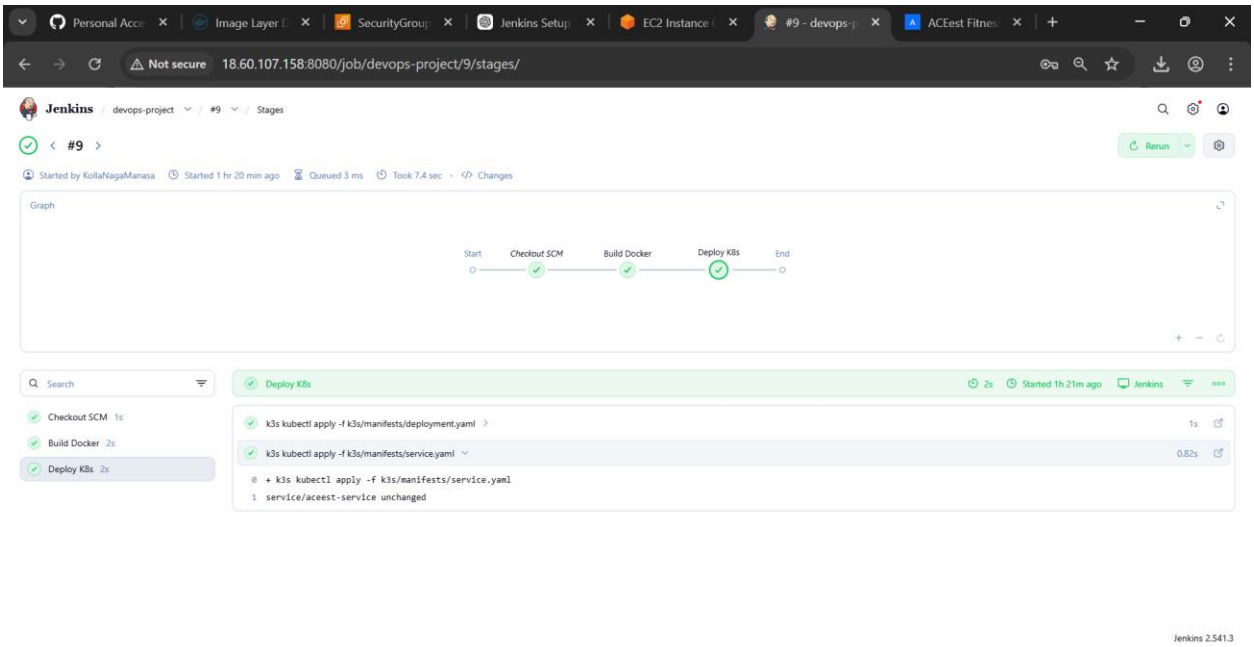
Personal Acce | Image Layer D | SecurityGroup | Jenkins Setup | EC2 Instance | Console - #9 | ACEest Fitnes

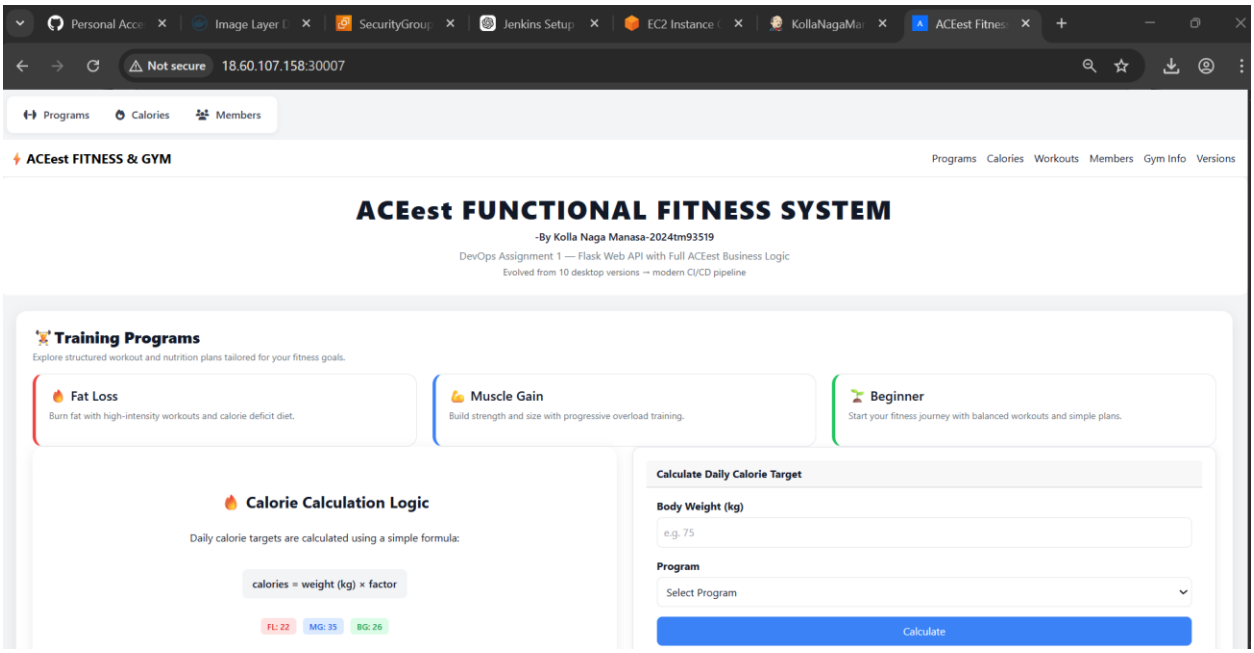
Not secure 18.60.107.158:8080/job/devops-project/9/console

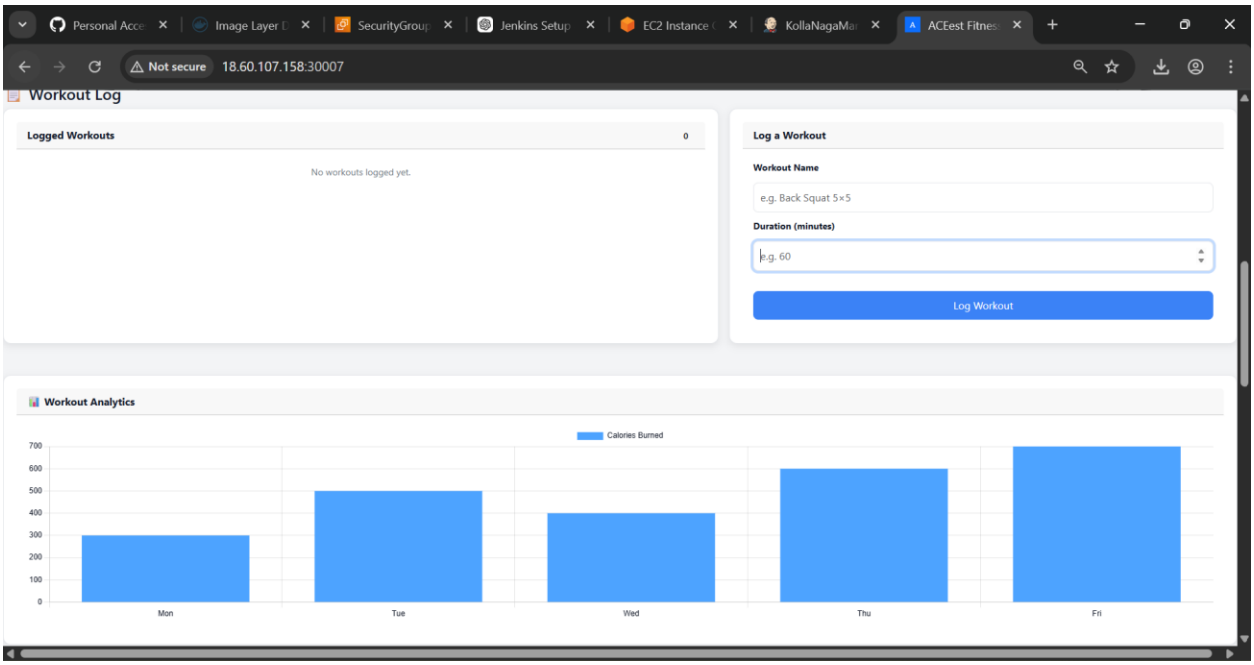
Jenkins / devops-project / #9

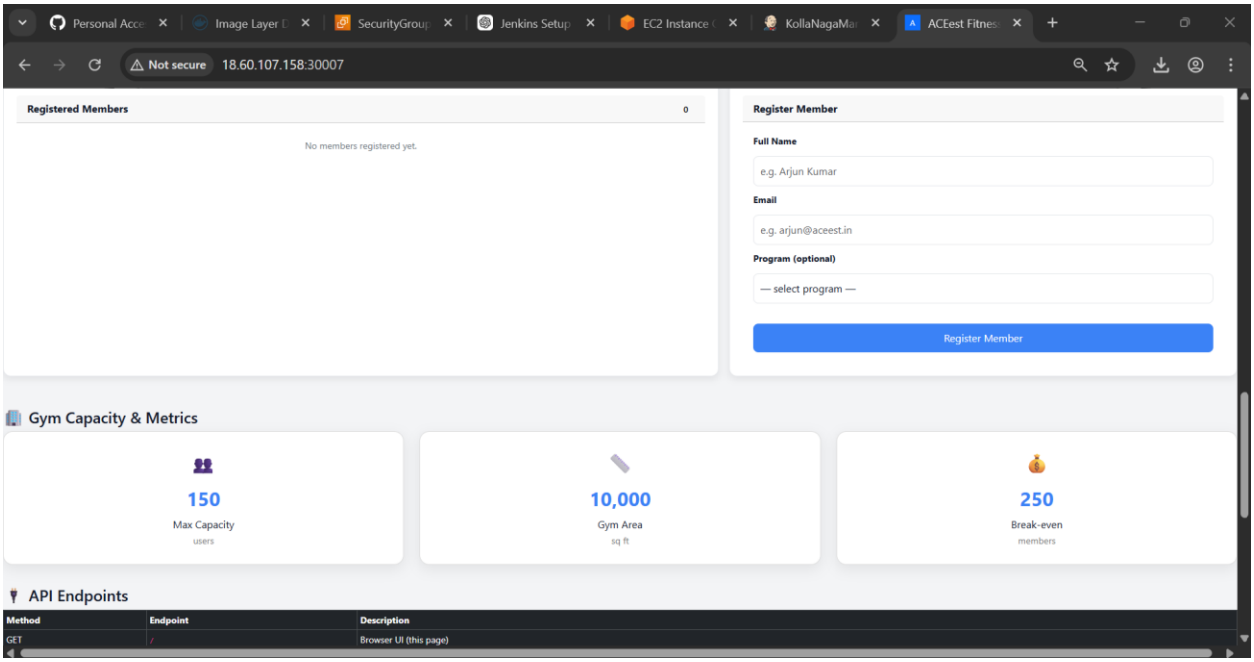
```
#9 [5/5] COPY . /app
#9 DONE 0.1s

#10 exporting to image
#10 exporting layers 0.0s done
#10 writing image sha256:7cf7af9fade8ebf2fc421118cf64846ffa1609746e710be4ecf26cbe65076d1e done
#10 naming to docker.io/library/aceest-app done
#10 DONE 0.1s
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy K8s)
[Pipeline] sh
+ k3s kubectl apply -f k3s/manifests/deployment.yaml
deployment.apps/aceest-deployment unchanged
[Pipeline] sh
+ k3s kubectl apply -f k3s/manifests/service.yaml
service/aceest-service unchanged
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

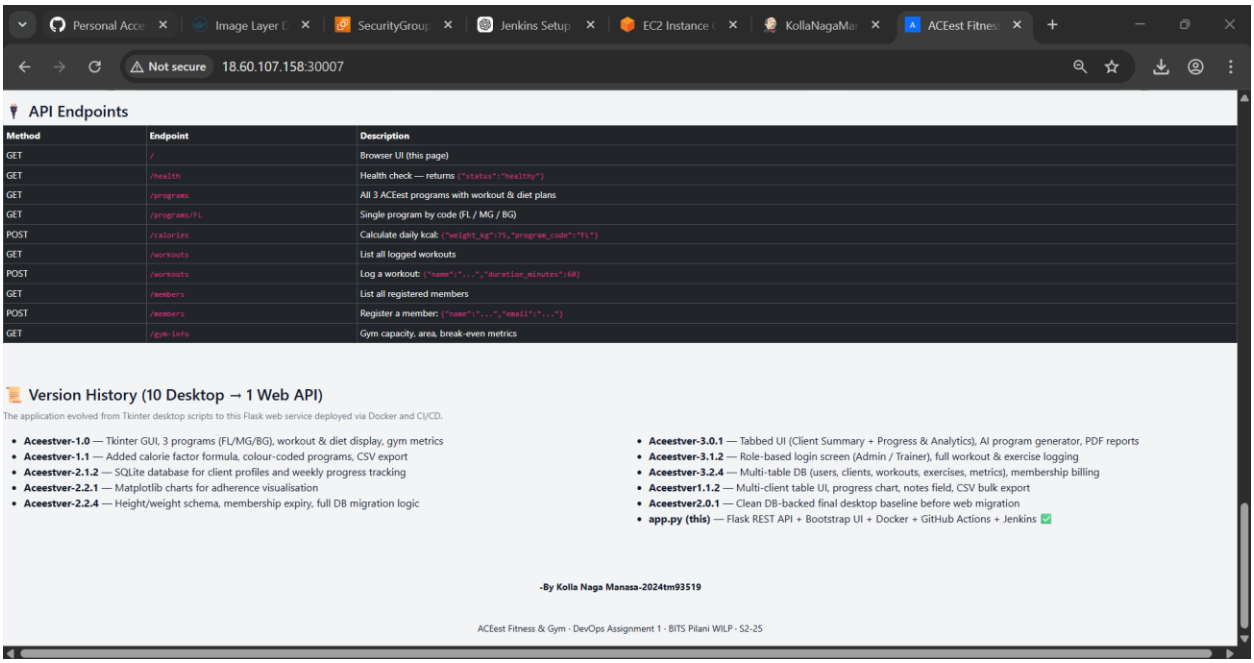
Jenkins 2.541.3

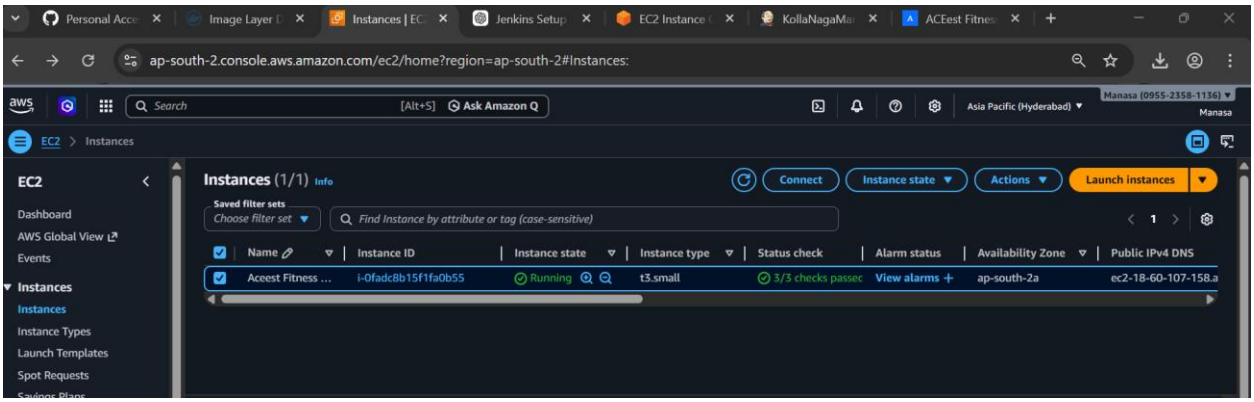

Not secure 18.60.107.158:8080/job/devops-project/9/stages/
Jenkins devops-project #9 Stages
#9
Rerun
Started by KollaNagaManasa Started 1 hr 20 min ago Queued 3 ms Took 7.4 sec Changes
Graph
Start Checkout SCM Build Docker Deploy K8s End
Search
Checkout SCM 1s
Build Docker 2s
Deploy K8s 2s
Deploy K8s 2s Started 1h 21m ago Jenkins
k3s kubectl apply -f k3s/manifests/deployment.yaml 1s
k3s kubectl apply -f k3s/manifests/service.yaml 0.82s
+ k3s kubectl apply -f k3s/manifests/service.yaml
service/aceest-service unchanged
Jenkins 2.541.3


Not secure 18.60.107.158:30007
Programs Calories Members
ACEest FITNESS & GYM
Programs Calories Workouts Members Gym Info Versions
ACEest FUNCTIONAL FITNESS SYSTEM
-By Kolla Naga Manasa-2024tm93519
DevOps Assignment 1 — Flask Web API with Full ACEest Business Logic
Evolved from 10 desktop versions → modern CI/CD pipeline
Training Programs
Explore structured workout and nutrition plans tailored for your fitness goals.
Fat Loss
Burn fat with high-intensity workouts and calorie deficit diet.
Muscle Gain
Build strength and size with progressive overload training.
Beginner
Start your fitness journey with balanced workouts and simple plans.
Calorie Calculation Logic
Daily calorie targets are calculated using a simple formula:
calories = weight (kg) × factor
FL: 22 MG: 35 BG: 26
Calculate Daily Calorie Target
Body Weight (kg)
e.g. 75
Program
Select Program
Calculate


Personal Acce
Image Layer D
SecurityGroup
Jenkins Setup
EC2 Instance
KollaNagaMa
ACEest Fitnes
Not secure
18.60.107.158:30007
Workout Log
Logged Workouts
0
No workouts logged yet.
Log a Workout
Workout Name
e.g. Back Squat 5×5
Duration (minutes)
e.g. 60
Log Workout
Workout Analytics
Calories Burned
700
600
500
400
300
200
100
0
Mon
Tue
Wed
Thu
Fri


Personal Acce
Image Layer D
SecurityGroup
Jenkins Setup
EC2 Instance
KollaNagaMa
ACEest Fitnes
Not secure
18.60.107.158:30007
Registered Members
0
No members registered yet.
Register Member
Full Name
e.g. Arjun Kumar
Email
e.g. arjun@aceest.in
Program (optional)
— select program —
Register Member
Gym Capacity & Metrics
150
Max Capacity
users
10,000
Gym Area
sq ft
250
Break-even
members
API Endpoints
Method
Endpoint
Description
GET
/
Browser UI (this page)


Personal Acce
Image Layer D
SecurityGroup
Jenkins Setup
EC2 Instance (
KollaNagaMa
ACEest Fitnes
Not secure
18.60.107.158:30007
API Endpoints
Method | Endpoint | Description
GET | / | Browser UI (this page)
GET | /health | Health check — returns {"status":"healthy"}
GET | /programs | All 3 ACEest programs with workout & diet plans
GET | /programs/FL | Single program by code (FL / MG / BG)
POST | /calories | Calculate daily kcal: {"weight_kg":75,"program_code":"FL"}
GET | /workouts | List all logged workouts
POST | /workouts | Log a workout: {"name":"...","duration_minutes":60}
GET | /members | List all registered members
POST | /members | Register a member: {"name":"...","email":"..."}
GET | /gym-info | Gym capacity, area, break-even metrics
Version History (10 Desktop → 1 Web API)
The application evolved from Tkinter desktop scripts to this Flask web service deployed via Docker and CI/CD.
Aceestver-1.0 — Tkinter GUI, 3 programs (FL/MG/BG), workout & diet display, gym metrics
Aceestver-1.1 — Added calorie factor formula, colour-coded programs, CSV export
Aceestver-2.1.2 — SQLite database for client profiles and weekly progress tracking
Aceestver-2.2.1 — Matplotlib charts for adherence visualisation
Aceestver-2.2.4 — Height/weight schema, membership expiry, full DB migration logic
Aceestver-3.0.1 — Tabbed UI (Client Summary + Progress & Analytics), AI program generator, PDF reports
Aceestver-3.1.2 — Role-based login screen (Admin / Trainer), full workout & exercise logging
Aceestver-3.2.4 — Multi-table DB (users, clients, workouts, exercises, metrics), membership billing
Aceestver1.1.2 — Multi-client table UI, progress chart, notes field, CSV bulk export
Aceestver2.0.1 — Clean DB-backed final desktop baseline before web migration
app.py (this) — Flask REST API + Bootstrap UI + Docker + GitHub Actions + Jenkins
-By Kolla Naga Manasa-2024tm93519
ACEest Fitness & Gym · DevOps Assignment 1 · BITS Pilani WILP · S2-25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Instances
Instance Types
Launch Templates
Spot Requests
Instances (1/1) Info
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Find instance by attribute or tag (case-sensitive)
Name | Instance ID | Instance state | Instance type | Status check | Alarm status | Availability Zone | Public IPv4 DNS
Aceest Fitness ... | i-0fadc8b15f1fa0b55 | Running | t3.small | 3/3 checks passed | View alarms | ap-south-2a | ec2-18-60-107-158.a